10. Show that the given points form a parallelogram :

$A(2.5, 3.5)$, $B(10, -4)$, $C(2.5, -2.5)$ and $D(-5, 5)$

11. If the points $A(2,2)$, $B(-2, -3)$, $C(1, -3)$ and $D(x, y)$ form a parallelogram then find the value of x and y.

12. Let $A(3, -4)$, $B(9, -4)$, $C(5, -7)$ and $D(7, -7)$. Show that $ABCD$ is a trapezium.

13. A quadrilateral has vertices at $A(-4, -2)$, $B(5, -1)$, $C(6, 5)$ and $D(-7, 6)$. Show that the mid-points of its sides form a parallelogram.

5.5 Straight Line

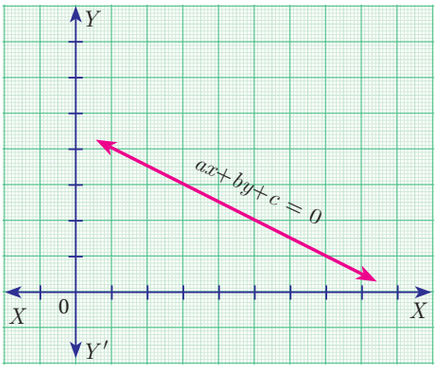


Fig. 5.26

Any first degree equation in two variables x and y of the form $ax + by + c = 0$ ...(1) where a, b, c are real numbers and at least one of a, b is non-zero is called "Straight line" in XY plane.

5.5.1 Equation of coordinate axes

The X axis and Y axis together are called coordinate axes. The x coordinate of every point on OY (Y axis) is 0. Therefore equation of OY (Y axis) is $x = 0$ (fig 5.27)

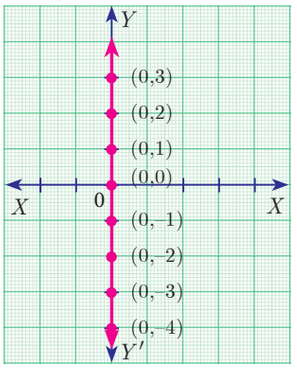


Fig. 5.27

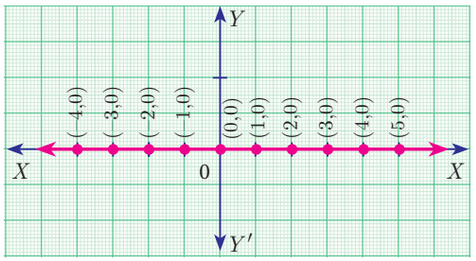


Fig. 5.28

The y coordinate of every point on OX (X axis) is 0. Therefore the equation of OX (X axis) is $y = 0$ (fig 5.28)

5.5.2 Equation of a straight line parallel to X axis

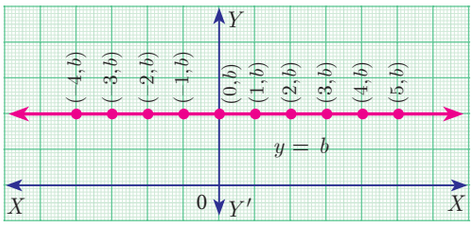


Fig. 5.29

Let AB be a straight line parallel to X axis, which is at a distance 'b'. Then y coordinate of every point on 'AB' is 'b'. (fig 5.29)

Therefore, the equation of AB is $y = b$

Note

- If $b > 0$, then the line $y=b$ lies above the X axis
- If $b < 0$, then the line $y=b$ lies below the X axis
- If $b = 0$, then the line $y=b$ is the X axis itself.

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